

Mahavitaran Voice — Test Call Latency & Usage Report

Three test calls (30 seconds / 2 minutes / 5 minutes) placed through Exotel to the deployed EC2 voice agent on 14 July 2026. This report gives the exact per-turn latency, and the Gemini token / Sarvam character usage, captured from the live systemd (journalctl) logs — nothing estimated.

1. Executive Summary

All three calls ran to completion with no hangups. Average end-to-end response latency stayed under the 2000 ms target in all three calls; a small number of individual turns (tool-call-heavy turns, or turns following a barge-in) spiked above it. The earlier "rings then hangs up after 1 second" issue was not observed in any of these three calls — all ran multiple full conversational turns.

Call	Duration	Turns	Barge-ins	Avg total latency	Max total latency	Result
Call 1 (30s test)	32s	2	0	1782 ms	1853 ms	Pass
Call 2 (2min test)	2m 11s	9	5	1577 ms	2254 ms	Pass*
Call 3 (5min test)	3m 42s+ (log window)	22	9	1396 ms	2039 ms	Pass*

*One individual turn in Call 2 and Call 3 exceeded the 2000 ms target (2254 ms and 2039 ms respectively) — both were multi-round turns where the model made a tool call (get_bill / register_complaint) requiring a second LLM round-trip before it could speak. The call average stayed well under target in every case.

Call 3's log capture began 10 minutes before the query was run, which landed mid-call (turn 7 onward) rather than at call start — turns 1–6 are not shown turn-by-turn below. The call-level summary and barge-in totals, however, are logged once at call teardown and cover the **full** 22-turn call regardless of when the log capture started.

2. What Each Latency Stage Means

Stage	Meaning
pipe	Internal handoff/queueing time between pipeline stages (VAD → STT → LLM → TTS). Normally single-digit to low double-digit ms.
stt	Sarvam Saaras speech-to-text round trip for the caller's utterance.
llm	Gemini 2.5 Flash streaming call: time to first token plus generation, per round. A turn needing a tool call (verify_consumer, get_bill, register_complaint, end_call) runs 2–3 LLM rounds, each logged separately.
tts	Sarvam Bulbul time to first audio chunk for the reply (streaming synthesis; cached lines return in ~1 ms).
total	"utterance → first-audio": wall-clock time from the caller finishing speaking to the first byte of the agent's spoken reply reaching the WebSocket. This is the number a caller actually perceives as "how long before it answered."
endpoint hangover	A deliberate 900 ms (early in a call) or 400 ms (once the caller is verified/known) silence-confirmation delay before the VAD decides the caller has finished talking. This is added on top of "total" and is a tunable design parameter, not network or processing latency.

3. Call 1 — 30 Second Test (session 1aa2ba45e99f)

19:36:08 – 19:36:40 UTC, 32 seconds, 2 completed turns, 0 barge-ins. Ended via Exotel stop event (rx media=1624, tx media=53).

Per-turn latency

Turn	Rounds	pipe (ms)	stt (ms)	llm (ms)	tts (ms)	total (ms)	hangover
1	1 (greeting only response)	14	81	1265	350	1710	+900
2	2 (tool call: verify_consumer)	25	80	1451 (avg of 2 rounds)	297	1853	+900

Call summary (logged by the server): pipe avg 20 ms · stt avg 81 ms · llm avg 1358 ms · tts avg 324 ms · **total avg 1782 ms, max 1853 ms** (target <2000 ms — pass).

Gemini token usage

LLM call	Prompt tokens	Completion tokens	Total tokens
1	5067	30	5097
2 round 0	5114	26	5140
2 round 1	5217	23	5240
TOTAL	15,398	79	15,477

Sarvam TTS character usage

313 characters requested total: 87 cached (greeting, 0 billed) + 226 live-synthesized and billed.

Note: STT usage logging (audio seconds → transcript characters) is wired into the REST fallback path (sarvam_stt.py) but this deployment uses the newer streaming STT path (sarvam_stt_stream.py), which does not yet emit that line — see Section 7.

4. Call 2 — 2 Minute Test (session 905331a4dbcb)

19:41:08 – 19:43:19 UTC, 2 min 11 s, 9 turns, 5 barge-ins (all during agent speech). Successful consumer verification for Sunita Deshmukh / 170023456789. Ended via Exotel stop event (rx media=6545, tx media=199).

Call summary (logged by the server): pipe avg 30 ms max 79 ms · stt avg 81 ms max 84 ms · **llm avg 1195 ms max 1847 ms** · tts avg 271 ms max 315 ms · **total avg 1577 ms, max 2254 ms** (target <2000 ms — pass on average).

Gemini token usage

14 LLM calls across 9 turns (5 turns needed a second round for a tool call). **Total: 74,425 prompt + 425 completion = 74,850 tokens.**

Sarvam TTS character usage

1,146 characters requested total: 105 cached (0 billed) + 1,041 live-synthesized and billed.

5. Call 3 — 5 Minute Test (session f784e82baa83)

Full call: 22 turns, 9 barge-ins (all during agent speech, 0 during "thinking", longest run of 4 consecutive barge-ins). Ended cleanly via the agent's own end_call tool after the caller said thank-you / goodbye — this call completed the full flow: identity verification, bill lookup (■3,134 for June 2026, 388 units), a high-bill complaint filed (SR2607EF9B69), and a new-connection request. Log capture began 10 minutes before the query and landed mid-call, so the turn-by-turn table below covers turns 7–22; the call-level summary line covers the full call.

Call summary (logged by the server, full call): pipe avg 21 ms max 52 ms · stt avg 81 ms max 82 ms · llm avg **1046 ms max 1657 ms** · tts avg 284 ms max 981 ms · **total avg 1396 ms, max 2039 ms** (target <2000 ms — pass on average). Barge-in summary: 9 total interruptions, all during agent speech, 0 suspected false positives, max 4 consecutive.

Per-turn latency (turns 7–22, as captured)

Turn	Rounds / notes	pipe	stt	llm	tts	total (ms)	hangover
7	1	16	80	1001	324	1421	+900
8	2 (verify_consumer, refused — bad digits)	32	80	1152	1 / 81 (r1)	1265	+900
9	2 (verify_consumer, refused — bad digits)	37	81	1143	305	1565	+900
10	2 (verify_consumer, refused — bad digits)	49	81	1345	1 / 36 (r1)	1476	+900
11	cancelled by barge-in before completion	—	—	—	—	—	—
12	1	21	81	1037	323	1462	+900
13	3 (verify_consumer ✓, get_bill, final)	52	81	1217	1	1350	+900
14	1 (short "Hmm" filler, anomalous)	12	80	0	981	309	+900
15	1	21	81	1000	317	1419	+400
16	2 (register_complaint)	10	81	1121	1	1212	+400
17	1	34	82	1112	292	1521	+400
18	1 (language switch hi→mr)	21	80	1200	346	1647	+400
19	1	11	80	665	309	1065	+400
20	1	11	81	843	290	1226	+400
21	1	8	81	992	317	1398	+400
22	2 (end_call)	21	81	1024	348	1474	+400

Gemini token usage (turns 7–22 as captured, 22 LLM calls)

Total: 120,740 prompt + 707 completion = 121,447 tokens for this visible portion of the call. Prompt size grows turn-over-turn (5386 → 5562) as conversation history accumulates in context — expected behaviour, not a leak.

Sarvam TTS character usage (turns 7–22 as captured)

1,906 characters requested total: 58 cached (0 billed) + 1,848 live-synthesized and billed.

6. Combined Totals — All Three Calls

Token/character totals for Call 3 reflect only the captured tail (turns 7–22); turns 1–6 are not included in these sums, so true full-call totals for Call 3 are somewhat higher than shown.

Call	LLM calls	Prompt tokens	Completion tokens	Total tokens	TTS characters
Call 1 (30s)	3	15,398	79	15,477	313 (87 cached)
Call 2 (2min)	14	74,425	425	74,850	1,146 (105 cached)
Call 3 (5min, captured tail)	22	120,740	707	121,447	1,906 (58 cached)
TOTAL	39	210,563	1,211	211,774	3,365 (250 cached)

7. Server Latency vs. Telephony Latency

What is measured (server-side, precise): every stage above — pipe, stt, llm, tts, and the resulting "utterance → first-audio" total — is timestamped inside the FastAPI process itself, from the moment our own VAD/endpointer decides the caller stopped talking, to the moment our own TTS layer hands the first PCM chunk to the outbound WebSocket. This is genuinely exact; it does not include any network transport time.

What is not separately measured (telephony / network leg): the time for that first audio chunk to actually travel EC2 → Exotel → the PSTN carrier → the caller's handset, and the equivalent return trip for the caller's voice. This journey adds real, additional delay on top of every "total" figure above, but none of it is currently timestamped in our logs — Exotel does not echo back delivery timestamps on the media WebSocket, and our code does not yet log its own send-time vs. Exotel's ack.

What can be said qualitatively: call audio was clean and responsive in all three tests with no reported echo, garble, or dead air beyond the measured server latency, and all three calls ran full multi-turn conversations without disconnecting — so telephony-leg latency and packet loss were not large enough to break the conversation flow. But an exact millisecond figure for the carrier/network leg cannot be produced from current instrumentation.

To measure it precisely, two additions would be needed: (1) log a timestamp when each outbound media WebSocket frame is sent and compare it against Exotel's own per-call quality report (available from the Exotel dashboard / API for each Call SID), or (2) round-trip a small heartbeat/echo probe over the same media path and time it. Happy to wire either of these in if precise telephony-leg latency becomes a requirement — for now the figures in this report are the true, exact server processing latency, which is also the only leg the codebase controls.

8. Notable Observations (Not Explicitly Requested, but Visible in the Data)

Observation	Detail
STT digit misrecognition	Calls 2 and 3 both show verify_consumer being refused multiple times because the Sarvam STT transcript of a spoken 12-digit consumer number came back with the wrong digit count or stray words mixed in (e.g. "1700 234 5 ■■■■ ■■■■ 789", "17002 2 ■■ ■■■■ ■■ ■■■■ 6,7,8,9"). The tool-refusal guardrail worked correctly (it never passed a bad number through), but the caller had to repeat the number 3–4 times in Call 3 before it verified. Worth checking if a slower confirm-by-digit flow, or DTMF capture instead of speech, would be more reliable for this specific field.
Barge-in speech_ms=0 on every interruption	All 14 barge-in events across Calls 2 and 3 logged speech_ms=0 at the moment of interruption. This is the value at the instant the barge-in fires, not a bug by itself, but worth a quick check that the barge-in threshold is timing genuine caller speech and not reacting to echo/bleed from the agent's own voice — 0 suspected false positives were flagged by the code's own detector, but that detector's confidence is worth spot-checking against the actual audio.
Turn 14 anomaly (Call 3)	Turn 14 (caller said "Hmm") shows llm=0 ms and a state-machine warning ("ILLEGAL transition speaking → thinking") immediately before it. Response was still delivered (981 ms tts) so the call was not affected, but this pattern is worth a look if it recurs — likely a very short filler utterance racing a state transition.